

Year 10 Metal — Program and Scope & Sequence

Teacher-developed, source-backed review draft · current course evidence with 2025 syllabus forward alignment

Stage 5 · Industrial Technology – Metal	Local review set v1.0	Prepared 5 August 2026
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Status: Local review document only. It is not NESA approved and does not change the live course, formal notifications or school program. Current local decisions and unavailable evidence remain Teacher to confirm.

Source authority

Source role	Exact basis used
Current course source	Local Yr-10-Metal checkout at commit 685e26def46c2aa4354db665fc2bbf71689efeb2; current source package checked by TASK 12B.
Course evidence	guided/data.js; SOURCES.md; YEAR-10-METAL-SOURCE-BOUNDARIES.md; ASSESSMENT-SOURCE-STATUS.md; verified plans; folio and task/exam pages.
Original programs	BBQ and Case and Folding Camping Shovel program/activity files control source content; historical 30% values do not override the 2026 schedule.
Assessment authority	Task 1 T2 W8 45%; Task 2 T3 W10 35%; Task 3 T4 W3 20%. Exact dates, outcomes, marks and approval remain teacher-controlled.
Current statutory position	Industrial Technology Years 7–10 Syllabus (2019) remains the current authority unless local early implementation is confirmed.
Forward-alignment authority	NSW Industrial Technology 7–10 Syllabus (2025): implementation from 2028; 2026–2027 are planning/preparation years. Official outcome/content pages and official Metal sample scope were checked.
Benchmark only	current-stage-5-timber-program.docx controls broad program structure and visual rhythm only; no Timber content or values were copied.

Whole-year scope overview

Term	Project/unit	Learning and evidence emphasis	Status
Term 1	BBQ and Case — guided modules 1–5	Brief/WMS; drawings; drilling; lathe planning; costing/fit-up	Current 18-week Project 1 spine
Term 2	BBQ and Case — guided modules 6–9	Industry/environment; finishing; case development; evaluation; Task 1 T2 W8	T2 W9–10 Teacher to confirm
Term 3	Folding Camping Shovel — guided modules 10–14	Drawings/WMS; head; fits/threads; milling/RPM; assembly/evaluation; Task 2 T3 W10	Current 10-week Project 2 spine
Term 4	Whole-course examination and consolidation	Revision; Task 3 T4 W3; post-assessment program	T4 W1–2 and W4–10 Teacher to confirm

Current guided-course spine

Module	Current weeks	Title	Practical/theory focus	Evidence or project position
1	W1–2	Brief, evidence and materials	Requirements, WMS, input–process–output.	P1 M1
2	W3–4	Drawings, rails and accurate setup	Stock forms, sketches, cutting lists and datums.	P1 M2
3	W5–6	Joining and controlled assembly	Hole transfer, drill-press SOP reasoning and inspection.	P1 M3
4	W7–8	WMS and repeatable work	Lathe planning, dimensions, surface quality and review.	P1 M4

5	W9–10	Costs, flowcharts and terminology	Plate/grill/guard planning, repeatability and fit-up.	P1 M5
6	W11–12	Industry and environmental responsibility	Stability, assembly, QA and industry/environment evidence.	P1 M6
7	W13–14	Preparation and finishes	Surface condition, finish choice, maintenance and impact.	P1 M7
8	W15–16	Case development and quality	Developments, bending control, hinges/handle evidence.	P1 M8
9	W17–18	Problem solving and final evidence	PMI, report evidence chain and final QA.	P1 M9
10	W1–2	Brief, modifications and job planning	Drawing system, WMS and impact inquiry.	P2 M1
11	W3–4	Risk, WMS and time/action planning	Head production, material response and quality readiness.	P2 M2
12	W5–6	Alloys and shovel-head production	Lathe SOP evidence, fits and thread geometry.	P2 M3
13	W7–8	Fits, threads and machining	Pivot/milling, RPM calculation and costing.	P2 M4
14	W9–10	Pivot, RPM, assembly and evaluation	Assembly gate, impact report, PMI and submission.	P2 M5

Detailed teaching sequence

Sequencing status: The content and evidence are source-grounded. Term/week placement is an editable planning allocation unless a current assessment week is stated. Exact lesson hours, interruptions and practical pace are Teacher to confirm.

Term 1 program

Timing	Content and learning focus	Practical/theory link	Student/teacher evidence	2025 outcomes	Control/gap
T1 W1–2	BBQ/Case M1 — brief, fixed requirements, evidence, WMS and input–process–output planning.	Establish source boundaries and quality hold points before practical production.	Requirements/evidence table; WMS explanation; verified flowchart branch.	SAF, COM, DES	Brief version, WMS template and first operations — Teacher to confirm
T1 W3–4	M2 — drawings, stock form, sketches, spreadsheet cutting list, datums, rails and end caps.	Use verified drawings and teacher-issued information before marking/cutting.	Material/stock justification; annotated sketch; cutting-list row; quality-gate checklist.	COM, DES, USE, INP, PRO	Materials, drawing revision and inspection criteria — Teacher to confirm
T1 W5–6	M3 — hole features, drill-press principles, controlled assembly and defect response.	Connect current SOP controls to drawing features and inspection evidence.	Transfer/check record; SOP annotation; drilled-feature inspection.	SAF, COM, USE, INP, PRO, EVL	Current SOP, controls, joining and rework approval — Teacher to confirm
T1 W7–8	M4 — WMS, supervised lathe work, reference surfaces, dimensions, surface quality and review.	Plan and document authorised machining through hold points and teacher review.	WMS-to-drawing trace; machining evidence trail; quality record.	SAF, COM, INP, PRO, EVL	Operation, tolerances, tools, setup and acceptance — Teacher to confirm
T1 W9–10	M5 — costs, flowcharts, terminology, plate/grill/guard planning and fit-up.	Use verified references, repeatability controls and dry/fit-up checks before permanent joining.	Hold-point plan; repeatability explanation; fit-up record; updated costing/sequence.	COM, DES, USE, INP, PRO	Sequence, jig/template, joining and cost data — Teacher to confirm

Term 2 program

Timing	Content and learning focus	Practical/theory link	Student/teacher evidence	2025 outcomes	Control/gap
T2 W1–2	M6 — structural/stability checks, industry/environment responsibility and quality assurance.	Restore source-backed collaboration, material-ordering, recycling/pollution and industry comparison if still taught.	Matched-component check; QA entry; group calculation/discussion record.	DES, ENV, INP, PRO, EVL, IND	Original activities and collaboration roles — Teacher to confirm
T2 W3–4	M7 — surface condition, preparation, finish performance, maintenance and service life.	Use current product/SDS information and teacher-specified finishing controls.	Before-condition record; cited finish justification; finish evaluation.	SAF, USE, INP, PRO, EVL, ENV	Finish, SDS, preparation, disposal and quality criteria — Teacher to confirm
T2 W5–6	M8 — case development, bending sequence, accumulated error, hinge/handle constraints.	Interpret the authorised development and evaluate concepts without inventing geometry.	Annotated development; forming hold point; decision rationale.	COM, DES, USE, INP, PRO, EVL	Geometry, equipment, sequence, hardware and flexibility — Teacher to confirm

T2 W7–8	M9 — problem solving, PMI, report evidence chain and final QA.	Diagnose issues and evaluate against criteria using authentic evidence.	Evidence-backed PMI; report conclusion; completed BBQ/Case and folio.	COM, DES, PRO, EVL	Task rubric, assessed outcomes and submission — Teacher to confirm
T2 W8	Formal Task 1: Project 1 and Folio — BBQ and Case (45%).	Teacher-observed practical evidence remains distinct from folio evidence.	Completed practical project, folio and observation record.	Approved task outcomes	Exact outcomes, marks, dates and approval — Teacher to confirm
T2 W9–10	Transition/catch-up period not defined in current guided source.	Do not invent a new project or sequence.	Teacher-directed evidence only.	Teacher to confirm	Authorised purpose and content — Teacher to confirm

Term 3 program

Timing	Content and learning focus	Practical/theory link	Student/teacher evidence	2025 outcomes	Control/gap
T3 W1–2	Shovel M1 — brief, drawing-system reading, modifications, job plan, WMS and impact inquiry.	Establish approvals and evidence before irreversible work.	Drawing relationship; WMS hold point; societal/environment questions.	SAF, COM, DES, USE, ENV	Drawings, modifications, WMS/job template and inquiry scope — Teacher to confirm
T3 W3–4	M2 — risk/time planning, shovel-head sequence, material behaviour and assembly-readiness checks.	Convert confirmed drawing features into controlled operations and checks.	Plan entry; material-response explanation; geometry/edge quality record.	SAF, COM, DES, INP, PRO, EVL	Forming method, time plan, criteria and adjustment authority — Teacher to confirm
T3 W5–6	M3 — alloys/materials, lathe SOP evidence, fits and thread geometry for authorised components.	Use source program and teacher-approved procedures; do not infer fit/thread specifications.	Material justification; SOP explanation; fit judgement; thread inspection.	SAF, COM, USE, INP, PRO	Materials, fits, threads, measuring tools and SOPs — Teacher to confirm
T3 W7–8	M4 — pivot centres, dry assembly, milling references, drilling RPM and costing.	Use teacher-supplied values and authorised settings; calculation is not machine permission.	Dry-assembly check; milling trace; labelled RPM calculation; costing sheet.	SAF, COM, DES, USE, INP, PRO	Settings, calculation values, speed selection and costs — Teacher to confirm
T3 W9–10	M5 — handle/pivot assembly, finish, impact report, careers/emerging technology, PMI and report.	Evaluate function and impacts with bounded evidence; restore career/industry activity if current.	Assembly gate; impact paragraph; career/CNC evidence; final PMI/report.	COM, DES, ENV, INP, PRO, EVL, IND	Career/enterprise evidence, finish and formal task details — Teacher to confirm
T3 W10	Formal Task 2: Project 2 and Folio — Folding Camping Shovel (35%).	Practical observation, project and folio evidence must remain distinct.	Completed project, folio and observation record.	Approved task outcomes	Exact outcomes, marks, dates and approval — Teacher to confirm

Term 4 program

Timing	Content and learning focus	Practical/theory link	Student/teacher evidence	2025 outcomes	Control/gap
T4 W1–2	Whole-course revision using drawings, measurement, WHS, materials, processes, quality, impacts and industry.	Review practical evidence separately from written/digital examination preparation.	Source-backed revision responses and evidence audit.	SAF, COM, USE, EVL, INP, ENV, IND	Revision program and any practical component — Teacher to confirm
T4 W3	Formal Task 3: Practical/theory examination (20%); current source states 60 minutes/70 marks.	Written evidence cannot by itself demonstrate INT5-PRO-01.	Submitted examination response.	Approved task outcomes	Outcome code set, session, access, submission and approval — Teacher to confirm
T4 W4–10	No authoritative local sequence found.	Do not invent catch-up, extension or an additional project.	Teacher-directed evidence only.	Teacher to confirm	Authorised purpose and sequence — Teacher to confirm

Outcome and evidence map

Implementation note: Formal 2026 assessment pages currently use the 2019 IND5 outcome bank. The INT5 mapping below is a forward-planning layer for the 2025 syllabus and must not replace formal codes until local implementation is authorised.

2025 outcome	Official wording	Coverage	Current evidence and required action
INT5-SAF-01	applies risk management and safe work practices in industrial technology contexts	Strong	WMS, risk/SOP evidence, equipment checks, practical observation and production records.

INT5-COM-01	communicates ideas and processes in industrial technology contexts	Strong/partial	Drawings, sketches, CAD, flowcharts, spreadsheets, folio/report; safe data management needs explicit evidence.
INT5-ENV-01	assesses the impact of industrial activities and practices on society and the environment	Partial/strong	Waste/recycling, industry pollution, finishes and shovel impact report; Cultural/ethical sourcing evidence remains a gap.
INT5-DES-01	selects and applies design, project management and technical skills in project production	Strong	Briefs, specifications, modifications, job/time plans, costing and quality hold points.
INT5-USE-01	selects and justifies the use of materials, tools and equipment to undertake projects	Strong with gaps	Stock forms, alloys, tools/processes and finishes; full material-content placement across Stage 5 needs confirmation.
INT5-EVL-01	evaluates the functional, aesthetic, economic and environmental qualities of products	Strong	Progressive checks, QA records, PMI and four-quality evaluation.
INT5-INP-01	analyses and applies a range of industrial processes in production contexts	Strong	Fabrication, sheet metal, lathe, milling, drilling, fits, threads, joining and finishing.
INT5-PRO-01	demonstrates practical skills in the production of projects	Strong if recorded	Two substantial practical projects; define observation checkpoints.
INT5-IND-01	investigates and explains a range of industry careers, enterprise opportunities and practices	Partial	Original programs include industry/CNC/careers; reintegration and enterprise evidence need confirmation.

Formal assessment and evidence register

Task	Current title/context	Timing	Weight	Evidence role	Teacher-controlled fields
Task 1	Project 1 and Folio — BBQ and Case	Term 2 Week 8	45%	Practical observation, project, planning, folio, quality and evaluation	Exact outcomes, rubric, marks, dates, submission and approval — Teacher to confirm
Task 2	Project 2 and Folio — Folding Camping Shovel	Term 3 Week 10	35%	Practical observation, machining/fabrication, folio, impact report and evaluation	Exact outcomes, rubric, marks, dates, submission and approval — Teacher to confirm
Task 3	Practical/theory examination	Term 4 Week 3	20%	Current source: 60 minutes/70 marks; drawings, WHS, measurement, processes and impacts	Outcome code set, practical component, session, access, submission and approval — Teacher to confirm

Teacher to confirm register

No.	Decision/evidence required	Status
1	Whether this is a standalone 100-hour course or one year of a 200-hour Stage 5 Metal elective.	Teacher to confirm
2	When the school/sector will adopt the 2025 syllabus and the formal reporting-code transition.	Teacher to confirm
3	Exact lesson allocation, calendar weeks, interruptions and practical pace.	Teacher to confirm
4	Current task notices: exact dates/times, outcomes, marks/rubrics, submission, approval and absence/misadventure details.	Teacher to confirm
5	Final briefs, drawings, materials, dimensions, tolerances, hardware, sequence, testing criteria and finishes.	Teacher to confirm
6	Local WHS: PPE, SOPs, SDSs, machinery access, supervision, risk assessments, settings and emergency procedures.	Teacher to confirm
7	Which specialist processes are practical, demonstrated, simulated/investigated or theory-only.	Teacher to confirm
8	The required collaborative activity, roles, common output and evidence of individual contribution.	Teacher to confirm
9	Authorised Aboriginal and Torres Strait Islander knowledge sources, Community protocols and ICIP boundaries.	Teacher to confirm
10	Required student-created graphical communication and applicable drawing/CAD expectations.	Teacher to confirm
11	Quantities/costing evidence, safe data-management practice and actual submission workflow.	Teacher to confirm

12	Careers/enterprise/industry evidence and the current source or industry link used.	Teacher to confirm
13	Accessibility adjustments and any separate Life Skills programming decisions.	Teacher to confirm
14	Which original-program activities remain in the compressed 18-week Project 1 and 10-week Project 2 sequence.	Teacher to confirm
15	Authorised use of Term 2 Weeks 9–10 and Term 4 Weeks 1–2 and 4–10.	Teacher to confirm

Program evaluation and approval

Field	Entry
Teacher/program author	Teacher to confirm
Faculty review date	Teacher to confirm
Syllabus implementation basis	Teacher to confirm
Evaluation notes and revisions	
Approval/version	Teacher to confirm